Progressive Proof Practice

MATH 147 | Lectures 7–8

Questions arranged by lecture. L1 = basics, L2 = standard theorem, L3 = application, L4 = challenge. CORE marks priority questions.

LECTURE 7 NOTES PP. 27–30

Convergence & Divergence

Q1. Uniqueness Lemma L1

Let $a, b \in \mathbb{R}$. Prove that if $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$.

Q2. Uniqueness of Limits L2 CORE

Prove that a sequence of real numbers has at most one limit.

Q3. A Non-Cauchy Sequence L2

Prove directly from the definition that

$$x_n = 1 + (-1)^n$$

is not Cauchy.

LECTURE 8 NOTES PP. 31–32

Algebra of Limits

Q4. Direct Limit Proof L1

Prove directly from the definition that

$$\lim_{n\to\infty} \frac{4n+3}{n+1} = 4.$$

Q5. Euler's Number L3

Let

$$a_n = \left(1 + \frac{1}{n}\right)^n.$$

Prove that $\{a_n\}$ is increasing and bounded above by 3.

Q6. Infinite Limit Application L4 CORE

Let $x_n = n^2/(1-n)$ for $n \geq 2$. Prove directly that $x_n \to -\infty$.